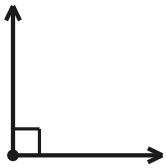
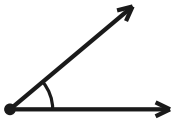
Angles

An angle is a turn between two rays. We measure it in degrees.

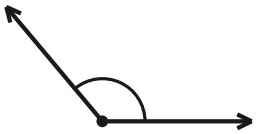
A full turn is 360 degrees. A square corner is 90.



right: 90°

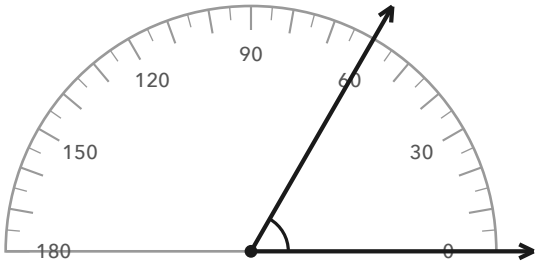


acute: under 90°



obtuse: over 90°

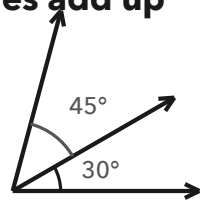
Read a protractor



1. Put the dot on the vertex.
2. Line up one ray with 0.
3. Read where the other ray crosses.

60 degrees

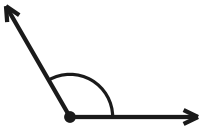
Angles add up



30 + 45 = [75] degrees

Two angles side by side: add their degrees.

Your turn!



acute right (obtuse)

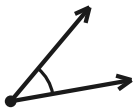
For teachers

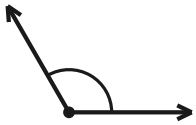
Goal: angles as turns measured in degrees, reading a protractor, and adding angles (4.MD.5–7). Compare with a square corner first: "Is it smaller than a corner, or bigger?"

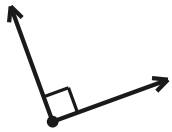
Angles: Practice

Square corner = 90 degrees. Straight line = 180 degrees.

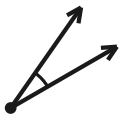
Part 1 Acute, right or obtuse? Compare with a square corner.

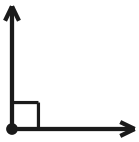
1)  **(acute)** right obtuse

2)  acute right **(obtuse)**

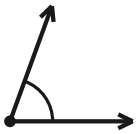
3)  acute **(right)** obtuse

4)  acute right **(obtuse)**

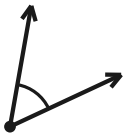
5)  **(acute)** right obtuse

6)  acute **(right)** obtuse

7)  acute right **(obtuse)**

8)  **(acute)** right obtuse

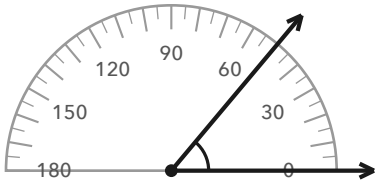
9)  acute right **(obtuse)**

10)  **(acute)** right obtuse

Angles: Practice (continued)

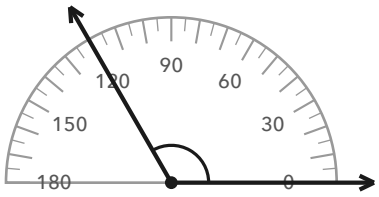
Part 2 Read the protractor How many degrees?

11)



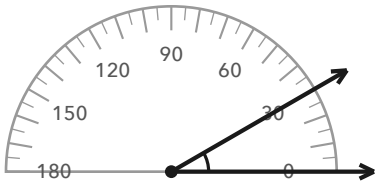
50 degrees

12)



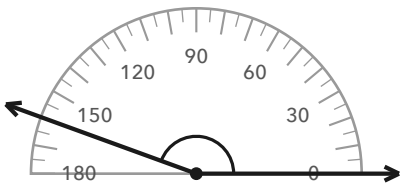
120 degrees

13)



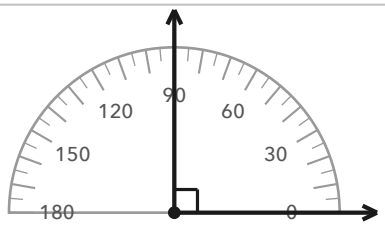
30 degrees

14)



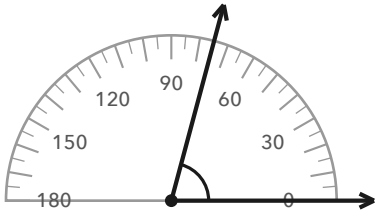
160 degrees

15)



90 degrees

16)

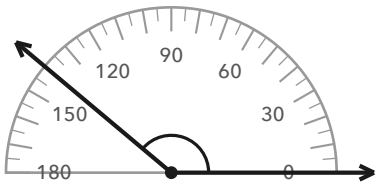


75 degrees

Angles: Practice (continued)

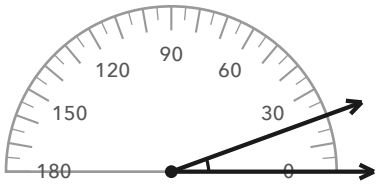
Part 2 Read the protractor (continued)

17)



140 degrees

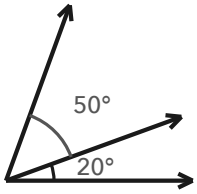
18)



20 degrees

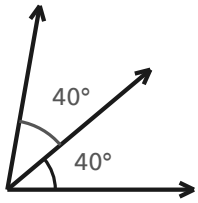
Part 3 Angles add up Add the two angles.

19)



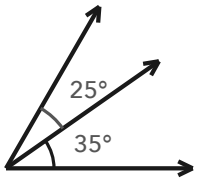
20 + 50 = 70 degrees

20)



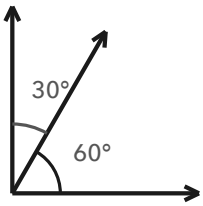
40 + 40 = 80 degrees

21)



35 + 25 = 60 degrees

22)



60 + 30 = 90 degrees

Part 4 Find the missing angle No help this time. Together they make the angle shown.

23) 35 + 55 = 90 degrees

24) 120 + 60 = 180 degrees

25) 45 + 45 = 90 degrees

26) 70 + 110 = 180 degrees

27) 25 + 65 = 90 degrees

28) 150 + 30 = 180 degrees

Angles: Practice (continued)

Part 4 Find the missing angle (continued)

29) 60 + 30 = 90 degrees

30) 95 + 85 = 180 degrees

Part 5 Add angles No help this time.

31) 45 + 30 = 75 degrees

32) 60 + 25 = 85 degrees

33) 90 + 45 = 135 degrees

34) 15 + 75 = 90 degrees

35) 100 + 50 = 150 degrees

36) 35 + 40 = 75 degrees

37) 70 + 80 = 150 degrees

38) 125 + 30 = 155 degrees

Angles: Practice (continued)

Part 6 Turns No help this time. A full turn is 360 degrees.

39) 1/4 turn = 90 degrees

40) 1/2 turn = 180 degrees

41) 3/4 turn = 270 degrees

42) 1 full turn = 360 degrees

43) 2 right angles = 180 degrees

44) 3 right angles = 270 degrees

Part 7 Make a right angle

45) 67 + 23 = 90 degrees

46) 43 + 47 = 90 degrees

47) 17 + 73 = 90 degrees

48) 58 + 32 = 90 degrees

49) 83 + 7 = 90 degrees

50) 32 + 58 = 90 degrees

Angles: Practice (continued)

Part 7 Make a right angle (continued)

51) 15 + 75 = 90 degrees

52) 46 + 44 = 90 degrees

53) 16 + 74 = 90 degrees

54) 6 + 84 = 90 degrees

55) 61 + 29 = 90 degrees

56) 52 + 38 = 90 degrees

57) 50 + 40 = 90 degrees

58) 38 + 52 = 90 degrees

59) 47 + 43 = 90 degrees

60) 22 + 68 = 90 degrees

61) 33 + 57 = 90 degrees

62) 85 + 5 = 90 degrees

Angles: Practice (continued)

Part 8 Make a straight angle

63) 86 + 94 = 180 degrees

64) 124 + 56 = 180 degrees

65) 103 + 77 = 180 degrees

66) 132 + 48 = 180 degrees

67) 159 + 21 = 180 degrees

68) 139 + 41 = 180 degrees

69) 160 + 20 = 180 degrees

70) 134 + 46 = 180 degrees

71) 137 + 43 = 180 degrees

72) 122 + 58 = 180 degrees

73) 77 + 103 = 180 degrees

74) 114 + 66 = 180 degrees

Angles: Practice (continued)

Part 8 Make a straight angle (continued)

75) $48 + \boxed{132} = 180$ degrees

76) $144 + \boxed{36} = 180$ degrees

77) $154 + \boxed{26} = 180$ degrees

78) $106 + \boxed{74} = 180$ degrees

79) $43 + \boxed{137} = 180$ degrees

80) $152 + \boxed{28} = 180$ degrees

Part 9 Remember? Problems from earlier packets.

81) area: $16 \times 9 = \boxed{144}$ sq ft

82) 6 m $= \boxed{600}$ cm

83) $1\frac{3}{8} + \frac{6}{8} = \boxed{2}\frac{\boxed{1}}{8}$

84) $0.56 \boxed{<} 0.6$

85) $\frac{9}{10} = \boxed{0.9}$

86) $1{,}206 \div 3 = \boxed{402}$

Angles: Practice (continued)

Part 10 Word problems Draw the angles.

87) A door opens 30 degrees, then 45 degrees more. How far has it opened?

Number sentence **30 + 45 = 75**

Answer **75** degrees

Draw or show your work

88) The hands of a clock make a square corner at 3:00. How many degrees is that?

Number sentence **a right angle = 90**

Answer **90** degrees

Draw or show your work

89) A skater turns half a circle. How many degrees does she turn?

Number sentence **1/2 of 360 = 180**

Answer **180** degrees

Draw or show your work

90) A pizza slice has a 60 degree corner. How many slices fit around the whole pizza?

Number sentence **360 ÷ 60 = 6**

Answer **6** slices

Draw or show your work

91) Challenge: Two angles make a straight line. One is 115 degrees. How big is the other?

Number sentence **180 − 115 = 65**

Answer **65** degrees

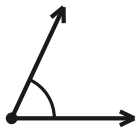
Draw or show your work

Check-In: Angles

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Acute, right or obtuse?

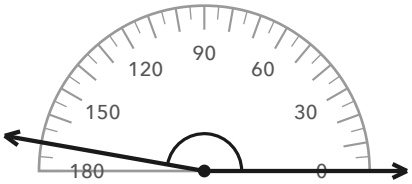
1) 

(acute) right obtuse

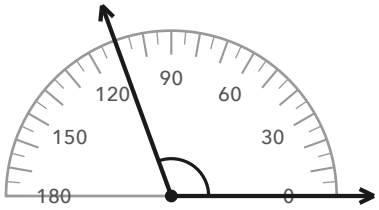
2) 

acute right (obtuse)

Part 2 Read the protractor

3) 

170 degrees

4) 

110 degrees

Part 3 Find the missing angle

5) 65 + **25** = 90 degrees

6) 75 + **105** = 180 degrees

7) 55 + **35** = 90 degrees

8) 55 + **125** = 180 degrees